

```

def merge_sort(arr):
    if len(arr) > 1:

        mid = len(arr) // 2

        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

        i = 0
        j = 0
        k = 0

        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1

        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

arr = list(map(int, input("Enter elements separated by space: ").split()))

print("Before sorting:", arr)

merge_sort(arr)

print("After sorting:", arr)
print("Time Complexity = O(n log n)")

```

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
==== RESTART: C:/Users/ROOPA SAI DIVYA/OneDrive/Documents/Big-O, Ω and Θ.py ====
Enter elements separated by space: 8 3 5 1 9 2
Before sorting: [8, 3, 5, 1, 9, 2]
After sorting: [1, 2, 3, 5, 8, 9]
Time Complexity = O(n log n)

>>>

```